

Full-production discretization P3: recorded CUDA matrix review

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement records and reviews the attached CUDA execution of the frozen P3 Stage-4 hardware/precision audit. Its purpose is to close the declared four-row CPU/CUDA complex128/complex64 matrix, not to promote the P3 claim or to generalise the result to every accelerator.

1 Recorded execution and source identity

The reviewed JSON reports PyTorch 2.13.0+cu130, CUDA 13.0, and one NVIDIA GeForce RTX 5070 Laptop GPU. Its canonical-backend hash is `e2b55702...56f70fc`; it equals both the frozen manifest authority and the current backend source. The current portable and hardware-audit source hashes also equal their frozen manifest entries. The artifact itself has SHA256 `b7590feb...1206a8d`.

2 Acceptance result

All six available assertions pass. The two CUDA rows are

configuration	energy error	limit	residual error	limit
complex128	$5.7489 \cdot 10^{-16}$	10^{-11}	$1.7181 \cdot 10^{-15}$	10^{-9}
complex64	$8.5245 \cdot 10^{-8}$	$5 \cdot 10^{-6}$	$3.7843 \cdot 10^{-7}$	$5 \cdot 10^{-5}$

Thus the audit verdict is `A3-FULL-HARDWARE-PRECISION-PASS`; the Stage-4 hardware gate is closed at its recorded-device, N8-field scope.

3 Devil’s-advocate and mandatory adversarial review

Sign and factor. The portable implementation is independently evaluated against the canonical CPU complex128 residual, so a sign or real-pairing factor mismatch would appear in the residual comparison. **Convention.** The evidence names the canonical backend hash and the review checks it against the frozen authority. **Units.** Both implementations use the fixed-torus P1 normalization and manifest thresholds. **Convergence.** This is a cross-device/precision agreement test, not a grid-convergence theorem; its N8 scope is explicit. **Hardcode masking.** The reviewed energy and residual errors are produced by the audit and checked against manifest limits, not copied into the code as targets. **Limit cases.** One GPU, two fields, and one grid do not prove all-device equivalence, a solution-ball error bound, or historical-solver continuum validity. External hardware reproduction remains welcome.

4 Remaining closure work

The independent analytic quadrature proxy and this hardware matrix now cover their executable P3 subtests. The load-bearing open passage is a uniform Galerkin/quadrature error estimate on P2 solution balls, followed by an integrated one-command verifier and an independent reproduction/tier review.

5 Reproduction

Run `python codes/foundations/a3_full_production_hardware_precision.py` in a CUDA-enabled Torch environment. Expected: 6/6 available assertions PASS and A3-FULL-HARDWARE-PRECISION-PASS.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Recorded CPU/CUDA complex128/64 precision matrix passes.
Scope:	Fixed T3, N8 homogeneous and Class-II stress fields, torch 2.13.0+cu130, one RTX 5070 Laptop GPU.
Dependencies:	Hash-pinned P1 backend and frozen P3 manifest.
Evidence grade:	EXECUTED; reviewed recorded-device evidence; T3 scaffold.
Reproduction command:	<code>python codes/foundations/a3_full_production_hardware_precision.py</code>
Expected output:	6/6 available assertions PASS; A3-FULL-HARDWARE-PRECISION-PASS.
Falsification gate:	A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after:	T3 / T3.
No-overclaim statement:	Not all-device equivalence, a uniform P2 bound, historical-solver result, T5, T6, or T7.
Next required action:	Uniform Galerkin error bound and integrated verifier.